SILAS OSUNBA

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PROFESSIONAL SUMMARY

Machine Learning Engineer with a background in Electronic and Computer Engineering and a track record of building production-grade AI systems. Experienced in architecting and deploying scalable ML solutions using Python, FastAPI, Docker, and AWS, with hands-on expertise in LLMs, RAG systems, and applied deep learning.

SKILLS

- **Programming:** *Python, SQL, JavaScript (React.js, Next.js), Java*
- **Machine Learning & MLOps:** *Scikit-learn, TensorFlow, Pandas, NumPy, SciPy, feature engineering, model evaluation, hyperparameter tuning, Hypothesis Experiment, A/B Testing*
- **ML Deployment and MLOps:** *FastAPI, Docker, MLflow, Model optimisation*
- **Data Stores:** *PostgreSQL, MySQL, MongoDB, Redis, AWS S3, Supabase, Vector Databases (ChromaDB, Pinecone, Faiss)*
- **Cloud & DevOps:** *AWS, Git, GitHub, CI/CD, GitHub Actions, Docker, Kubernetes*
- **Generative AI Frameworks:** *LangChain, CrewAI, Pydantic AI, LlamaIndex*
- **Automation:** *Make.com, n8n*
- **Soft skills:** *Problem Solving, Attention to detail, Effective communication, Leadership, Creative collaboration, Team work*
- **Professional:** *Technical Writing, Research, Microsoft Office Suite*

PROFESSIONAL EXPERIENCE

Condere Corp Ltd — England, United Kingdom

AI & Product Engineer *Jun. 2026 - Present*

Skills: Python, JavaScript, AWS, Git, GitHub, Docker, CI/CD

- *Architected and developed an LLM-augmented candidate recommendation and data-reuse platform using FastAPI, Next.js, and Supabase, transforming more than 7,432 historical submissions and 5,223 expert profiles into a searchable talent intelligence system.*
- *Designing agentic AI and human-in-the-loop systems for elimination of 80% of manual recruitment processes.*
- *Built and deployed an outreach agent that connects and integrates internal data sources, tools, enrichment platform and outreach messaging; reducing manual outreach processes by 95% using REST APIs and MCPs to improve data movement and operational workflow consistency.*

Moniepoint Inc. — Lagos, Nigeria

Software Engineering Fellow *Feb. 2026 - May 2026*

Skills: Java, JavaScript, Python, AWS, Git, Postman, Docker, Kubernetes, CI/CD, GitHub Actions

- *Designed and developed an analytics endpoints that queried merchant activity data to generate business insights for the Data Analytics and Growth teams.*
- *Deployed a containerized system that delivered business insights 75% faster and at 25% lower cost than previously deployed endpoints.*
- *Implemented an idempotent CSV-to-PostgreSQL ingestion pipeline using field-level validation, 5,000-record batch inserts, and ON CONFLICT DO NOTHING to reduce database round trips while safely handling invalid and duplicate events.*

Engineering.Fit() Research Laboratory — Lagos, Nigeria

Machine Learning Researcher *September 2024 - December 2025*

Skills: Python, Pytorch, Tensorflow, Scikit-Learn, Git, Microsoft Office Suite, Matplotlib, Seaborn, PowerBI

- *Developed and evaluated a CNN-based oil-spill detection system optimised for real-time inference on resource-constrained edge devices.*
- *Fine-tuned pretrained computer-vision models on a specialised oil-spill dataset and achieved 97% detection accuracy.*
- *Applied float16 quantization to reduce model size by 70% while ensuring predictive performance is preserved and improved deployment suitability for low-resource environments.*

EDUCATION

Bachelor of Science (B.Sc.), Electronic & Computer Engineering — *Sept. 2020 - Oct. 2025*

Lagos State University, Lagos, Nigeria

- **Cumulative GPA:** *4.77/5.0 (First Class Honors, Best of Graduating Class '25)*
- **Relevant Coursework:** *Research, Machine Learning & Neural Networks, Deep Learning, Software Engineering, Statistics and Probability, Stochastic Calculus, Computations and Programming, Numerical and Computational Analysis, Database Management & Optimization*

RESEARCH AND TECHNICAL PROJECT EXPERIENCE

QueryMate (Source Code)

Description: *An open-source Python package for querying large relational database systems using pure natural language. It provides natural language querying capabilities that translate questions into SQL and return responses in human-readable formats, with effective memory engineering implemented and designed with security in mind. Since relational database systems contain sensitive information that must be protected, rigorous security measures have been applied accordingly. This open-source project was created from observation that the current SQL toolkits of popular Agentic AI Frameworks - LangChain, LlamaIndex, and CrewAI - lack secure interaction with relational databases, which exposes these database systems to risk or attacks. Ultimately, this work/product was created specifically to eliminate the technical expertise typically required between non-technical teams and their databases. Designed the package for integration into analytics and self-service data applications.*

Automated Detection of Diabetic Retinopathy (Presentation, Live App)

Description: *Manual diagnosis of Diabetic Retinopathy (DR) is time-consuming, expensive, susceptible to human experts' subjectiveness, and limited by a shortage of ophthalmologists. This work addresses these challenges by leveraging machine learning and computer vision to automate DR detection. A CNN architecture model analyzes retinal fundus images achieving 96% recall while ensuring early and accurate diagnosis. The CNN architecture was inspired by ResNet and MobileNet.*

- *I deployed as a web-based application, enabling users to upload single or batch retinal fundus images for real-time DR analysis. The platform features secure patient data management, allowing storage, retrieval, tracking of predictions, and downloadable reports in PDF format.*
- *I presented this research at the Data Science and Medical Image Analysis Training in Nigeria (DATICAN 1.0), held in Lagos, Nigeria and secured the 3rd Runner-Up position.*

End-to-End Machine Learning System for Loan Eligibility Prediction (Code, Live App)

- *I developed a loan eligibility prediction system utilizing five machine learning algorithms - SVC, KNN, Logistic Regression, XGBoost, and Random Forest. Random Forest algorithm outperformed other ML algorithms across all business evaluation metrics, and hyperparameter tuning techniques were later applied to Random Forest to optimize its performance.*
- *I exposed the model for future loan eligibility predictions as REST API endpoints using FastAPI and designed the web application frontend with HTML and Bootstrap CSS for an intuitive user experience.*
- *Deployed the application to a cloud environment and implemented reusable preprocessing and prediction workflows.*

AWARDS & HONORS

- **Chancellor's/Dean's Award for the Engineering Valedictorian** – *Lagos State University, August 2026*
- **First E&P Scholar** – *First Exploration and Petroleum Development Company, August 2024.*
- **Undergraduate Scholarship Award for Excellent Performance in Engineering** – *Society of Petroleum Engineers International (SPEI), Lagos Section 61, July 2023.*
- **Vice Chancellor's Letters of Commendation/Award of Scholarship** – *Lagos State University Management and Senate, For an outstanding performance in the 2020/2021 academic session , September 2022.*